

Demonstrating TactorBots: A Haptic Design Toolkit for Exploration of Emotional Robotic Touch

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Figure 1: TactorBots: Haptic design toolkit for exploring emotional robotic touch (a) Hardware modules are worn on the forearm (b) It can be packaged in a compact case (c) It contains hardware modules and a web-based authoring tool

ABSTRACT

Emerging research has demonstrated the viability of emotional communication through haptic technology inspired by interpersonal touch. However, the meaning-making of artificial touch remains ambiguous and contextual. We see this ambiguity caused by robotic touch's "otherness" as an opportunity for exploring alternatives. To empower designers to explore emotional robotic touch, we devise TactorBots. It contains eight plug-and-play wearable tactor modules that render a series of social gestures driven by servo motors. Our specialized web GUI allows easy control, modification, and storage of tactile patterns to support fast prototyping. Taking emotional haptics as a "design canvas" with broad opportunities, TactorBots is the first "playground" for designers to try out various tactile sensations, feel around the nuanced connection between touches and emotions, and come up with creative imaginations.

CCS CONCEPTS

• Human-centered computing → Systems and tools for interaction design; Haptic devices.

KEYWORDS

Haptic Design Toolkit, Wearable, Emotional Robotic Touch, Design Research, Creativity Support

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1 INTRODUCTION

Touch, the very first sense to develop in the womb [5], is the primary sense for humans to explore the physical world and is integral to our everyday lives. It is also an effective communication channel, as studies have demonstrated that humans can convey emotions solely through touch [11, 12]. These findings have sparked an increased interest in replicating *familiar* social gestures for computer-mediated emotional communication [13]. However, since the meaning-making of artificial touch [19] is contextual and subjective, it is challenging to arrive at a generalizable haptic language

[7, 15]. On the other hand, researchers are reconsidering people's willingness to replace interpersonal physical contact [14, 20], and replicating realistic human affective touch with haptic devices may even fall into the uncanny valley [3, 18]. Under this circumstance, we start to ask, what alternatives can we explore?

A recent manifesto proposed by Jewitt et al. highlights the significance of exploring the richness and meaningfulness of social haptics and fostering interdisciplinarity in touch design beyond technology-driven development [14]. Resonating with it, our previous work, EmotiTactor [23], investigated how designers approach emotional haptics, which revealed the emergence of alternative metaphors beyond replicating human behaviors. We tentatively hypothesized that those metaphors could be evoked by robotic touch's *otherness*. *Otherness*, here, is an aesthetic category [4] for describing robot design that exceeds the dominant anthropomorphic or zoomorphic paradigms in their outward form [1], whose inherent ambiguity allows a broader spectrum of possible social responses to robot's behavior which benefits long-term engagement [2, 10, 21]. As extensively discussed in human-robot interaction (HRI), our work plans to intentionally emphasize the *otherness* feature in emotional robotic touch and investigate its impact in depth.

Inspired by the aforementioned findings, we propose treating emotional robotic touch as a "design canvas" that opens up space for alternative opportunities. As *ambiguity* can be utilized to encourage deeper interaction and more personal relationships when making sense [9], we emphasize the importance of embracing *ambiguity* in emotional haptic design by leveraging the balance between *the familiarity of interaction - recognizable social gestures* and *the otherness of form - robotic tactor's appearance and texture* [21].

To empower designers in exploring ambiguity-provoked alternatives, a haptic toolkit ready for independent operation in the wild is needed. Thus, we developed TactorBots [24], an open-source and open-hardware design toolkit. It contains 8 plug-and-play wearable modules that render 9 social gestures with servo-driven feedback. Having the design principle of *otherness* in mind, we devise each module in a minimalist form that follows its purpose function [6]. The small-scale modules can be worn on various body parts, while we take the forearm as the default due to social acceptability [22]. TactorBots' web-based software allows easy control, modification, and storage of tactile behaviors to support fast prototyping.

2 DESIGN OF TACTORBOTS

TactorBots is a design toolkit for exploring emotional robotic touch outside the lab. We followed the principles of designing toolkits for wearable embodiment interaction summarized by Ledo et al. [16] and Lei et al. [17] and set our design goals as Accessibility, Flexibility, and Trust. Moreover, we regard our toolkit as a probe for supporting evocative tasks and eliciting inspirational responses [8]. To provoke alternative interpretations, another key design goal is producing ambiguity by leveraging the *familiarity* and *otherness* of the robotic touch. Specifically, we make robotic interfaces perform distinguishable gestures while remaining true to their mechanistic nature in appearance and texture. This section presents the design process, features, and user flow of TactorBots platform.

2.1 TactorBots Hardware Modules

TactorBots hardware is designed in a modular, wearable fashion (Fig. 2). Each small-scale (8 x 5 x 4.5 cm), lightweight (72 g), and low-cost (< \$15 per Module) wearable module comprises three main parts: Case, Cover, and Tactor. To preserve the *otherness* feature, tactor prototypes were designed in a simple minimalist form that was shaped purely based on their purpose functions and fabricated by 3D printing. In Figure 2, we take the Shake Module as an example. The internal Tactor's shape and mechanism (Fig. 2b(5)) are designed to perform the target gesture in a minimalist form. The Tactors are driven by AGFrc B11DLS micro servo (180 degrees, 2.8 kg-cm) except Stroke Tactor, which uses an FS90R continuous rotation servo (360 degrees, 1.5 kg-cm). While having a consistent enclosure, each case and cover are adaptively designed to allow the Tactor's smooth movement while fixing the servo's placement. We restricted the Tactors' motion range in the software to prevent over-rotation for safety. An open entrance and small door (Fig. 2b(3)) are designed for assembling Tactor to the servo shaft easily. The enclosure and Tactor's outer edges are filleted for comfort and the downward surface fits the general-size limb shape with an attached adjustable velcro strap. The modules are designed using Fusion 360 and printed in PLA plastic. We make the cover clear while giving the Tactor a distinct orange to make the mechanisms visible, which supports sense-making. We open source¹ all the main parts in STL format with suggested printing settings and provide the hardware and accessories information in the Bill of Materials form (BOM). TactorBots also contains a Home Module hosting a custom control board, which bridges the Tactor Modules and PC. Tactor Modules

¹<https://www.ranzhourobot.com/tactorbots/>

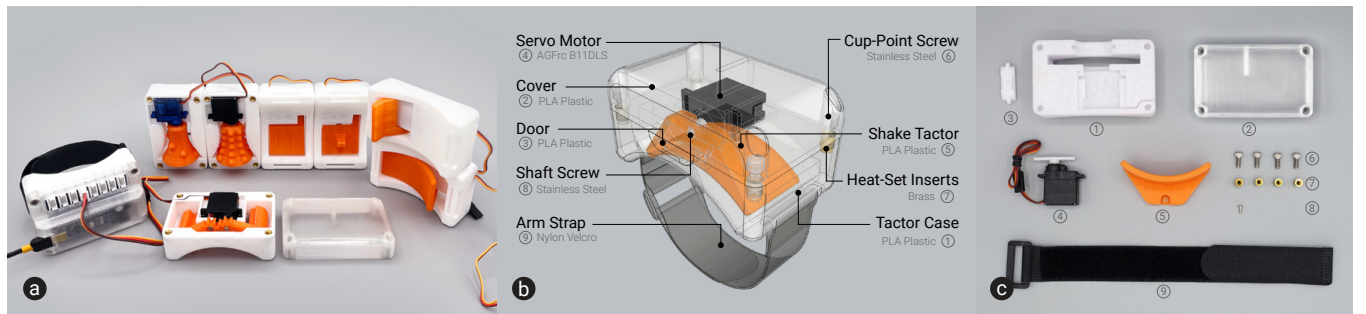
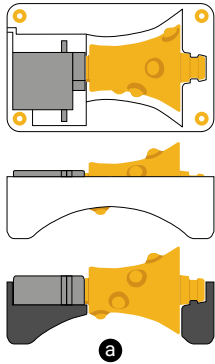
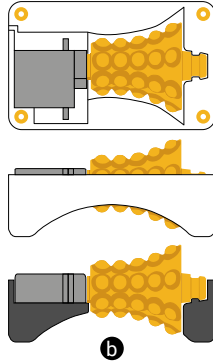


Figure 2: (a) TactorBots hardware modules (b) Mechanism of Shake Tactor (c) Components to make a Shake Tactor

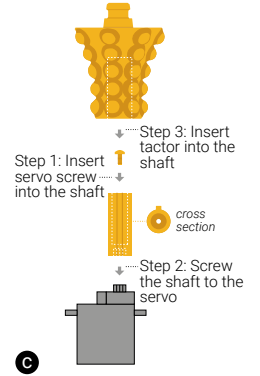
Rub and Stroke Modules



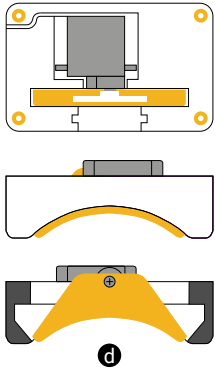
Stroke Module: As the stroke early prototype cannot fit into the new form factor, we design a novel fashion: a contoured barrel that rotates on an axis, with raised dots that are placed in a way that the point of contact changes as the device rotates. Users will feel continuous tactile stimulation as we use a 360 servo (FS90R) and notice that the sensation moves across the arm, which can mimic the stroking motion in a limited range.



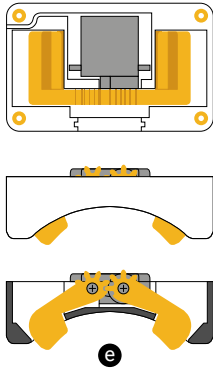
Rub Module: Similar to the Stroke Tactor's design while using a 180-degree servo. It applies subtle yet noticeable pressure on the skin when rotating back and forth. After several rounds of iteration, the current tactor has raised dots evenly distributed on the contoured barrel. The size of the dots is carefully designed to be not too small to be tickling but not too large to be stretching.



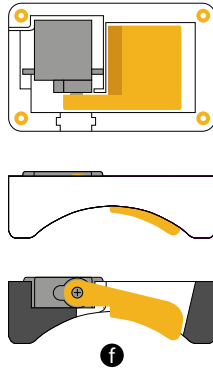
Force Feedback Modules



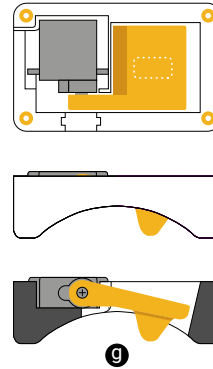
Shake Module: the tactor is designed in a concavity shape. When triggered, it will swing while making the arm sway.



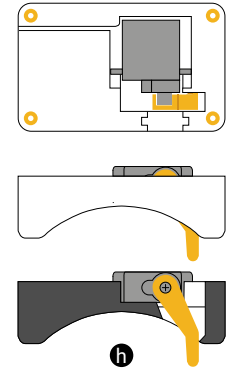
Squeeze Module: the tactor has two arms that can apply pressure from the lateral sides. We design a gear set to simplify early prototypes.



Pat/Hit Module: it creates large skin deformation with pressing or slapping. The curved surface fits better to limb's shape for full contact.



Tap Module: for a smaller contact point, we add a downward triangle shape, which helps it differentiate from the Pat/Hit Tactor.



Push Module: tactor sticks out of the case with 2cm and provides normal force at the lateral side.

Figure 3: Design of the Tactor Modules

are attached to Home Module with JST connectors, which are stable when plugged in and appropriate for wearable haptic applications with large-range body movements. Figure 3 presents the design of each Tactor Module. In our pre-assembled kits for following design exploration, we also provide the extension strap for fixing the Modules on different body parts and a pry tool to support unplugging the JST connectors.

2.2 User Flow and TactorBots Software

To operate TactorBots, the designer first opens the TactorBots web-based authoring tool in Google Chrome and attaches the Home Module to the PC with a micro USB cable (Fig. 4 Step 1). They then follow the instructions in the **Connection** (Fig. 4d) tab to connect it through Web Serial (Step 2). The next step is to proceed to the **Tactor Information** tab (Fig. 4b), which depicts the form factor, the recommended placement based on our formative study, and potentially associated emotions according to prior research [23]. When the designer decides to use a specific Tactor, they click the Activate button. An LED on the Home Module will light up to indicate the corresponding port. After attaching the Tactor to the

port using a servo wire with a JST connector, the user can wear the Module and proceed to the **Touch Generator** (Fig. 4c) to design the haptic cues. In this tab, six sliders control the Tactor behavior. For instance, the designer can set a minimum and maximum angle in *Touch Range* and then use *Time of Touching* to control how long the Tactor moves from the maximum to the minimum range. See [24] for more details about the parameters. When satisfied, they can name the pattern and save it to the **Archive** (Fig. 4e), where stored patterns can be played back and exported. The users' information and design data are stored in our backend for research analysis.

3 CONCLUSION

We present TactorBots, an open-hardware and open-source haptic toolkit for designing emotional robotic touch. We aim to expand the design space by leveraging the *ambiguity* and *otherness* of the robotic touch to provoke alternative interpretations and new design opportunities. Inspired by psychological findings on interpersonal emotional interactions, we employ gesture-based haptic technology as the starting point. To keep the *otherness* feature, we design the tactor modules with a minimalist form, which are shaped purely

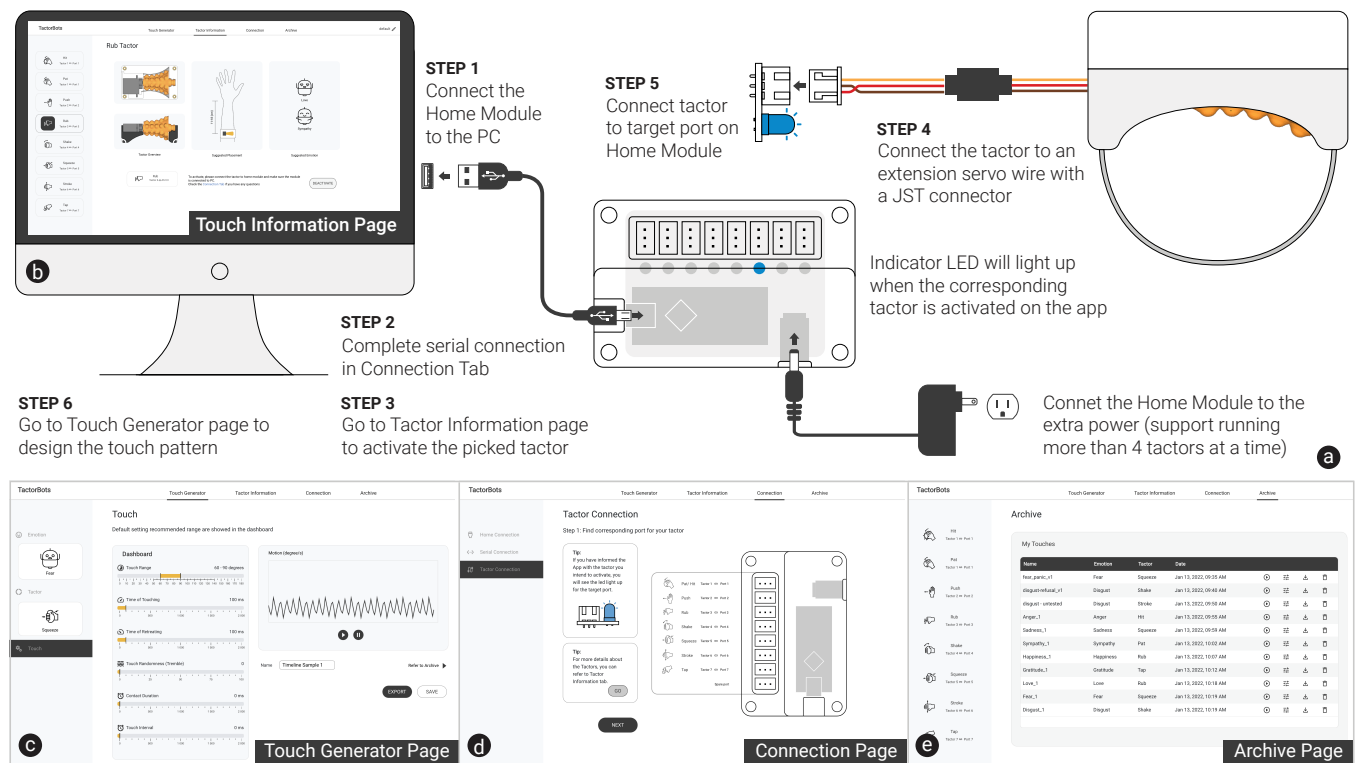


Figure 4: (a) User flow of operating TactorBots Toolkit. Main tabs on GUI authoring tool: (b) Touch information (c) Touch generator (d) Connection (e) Archive

based on their purpose functions. We demonstrate TactorBots to showcase how it could provoke reflective perception, alternative interpretations, and creative imaginations.

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